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Procedia CIRP 130 (2024) 1117–1123

www.elsevier.com/locate/procedia

57th CIRP Conference on Manufacturing Systems 2024 (CMS 2024)

Data-Driven Approach for Decision-Making in Reactive Disassembly Planning to Enable Case-Based Reasoning

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Abstract

Environmental issues require sustainable manufacturing, which includes improving the circularity of products. The inefficiency of product disassembly is a crucial barrier to industrial-scale circularity. One cause of inefficiency is using deterministic disassembly plans despite the uncertainties of end-of-life products, such as their varying conditions. Instead, disassembly requires the online adaptation of these plans whenever new information deviates from prior assumptions. This is known as reactive disassembly planning. Reactive disassembly planning faces individual and dynamic planning problems that depend on the state of the product, the disassembly process, and the resources in the disassembly system. Decisions in reactive disassembly planning affect the behavior and output of the disassembly system, which influences the profitability and sustainability of the disassembly. Due to this complexity, decision-making in reactive disassembly planning should be supported methodically. This contribution presents a data-driven approach supporting decision-making in reactive disassembly planning. It allows for the automatic adaptation of disassembly plans in response to plan deviations while considering the impact on economic and environmental metrics. Methods for solution space management, design of experiments, and material flow simulation are used to identify and select solution alternatives for unknown deviations. Selected solution alternatives are linked with their corresponding deviations and stored as cases in a case base. The case base enables the application of case-based reasoning, which utilizes the case base without having to identify and select solution alternatives. Therefore, it enables learning from experience and accelerates reactions to recurring plan deviations. In this contribution, the modules and the procedure of the approach are presented.

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Peer-review under responsibility of the scientific committee of the 57th CIRP Conference on Manufacturing Systems 2024 (CMS 2024)

Keywords: disassembly planning; decision-making; simulation; case-based reasoning

1. Introduction

Climate change, the scarcity of critical resources, and increasing waste pressure manufacturing companies to become more sustainable [1, 2]. One field of action is improving the circularity of products to reduce the product-specific resource and energy input, waste, and emissions [3]. The disassembly of end-of-life products (EOL products) is the first step in most circularity strategies and plays a crucial role in the transition to a circular economy [4]. However, the complexity, the high cost, and the time-consuming nature of disassembly remain barriers to this transition [5, 6]. Improving disassembly planning is essential to achieving a more economically and environmentally efficient disassembly [7, 8]. According to [9], disassembly planning involves making decisions based on information obtained from the inspection and sorting of EOL products. This information is used in disassembly planning to determine, organize, and manage disassembly activities, such as selecting the components to be disassembled. While predictive disassembly planning considers a generic product to determine an initial disassembly plan based on assumptions, reactive disassembly planning adapts this plan during the disassembly process whenever new information becomes available that deviates from the prior assumptions [10]. Reactive disassembly planning is necessary because disassembly is characterized by uncertainty. This leads to a changing information base that causes deviations from the disassembly plans [8, 9]. Deviations can be disposition-related (e.g., order cancellations), personnel-related (e.g., human errors), equipment-related (e.g., machine failure), or product-related (e.g., inseparable components) [11, 12]. While comparable deviations can occur in conventional production systems for the first three types, product-related deviations are disassembly-specific [11]. For example, unexpectedly inseparable compo-

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Peer-review under responsibility of the scientific committee of the 57th CIRP Conference on Manufacturing Systems 2024 (CMS 2024)
10.1016/j.procir.2024.10.215

nents can affect the number, sequence, and duration of process steps, resulting in a fundamentally different disassembly process and output [13]. Such deviations make deterministic plans infeasible or inefficient [8, 14]. Because of these characteristics, conventional production planning methods cannot be applied to reactive disassembly planning [8, 12, 14]. Instead, dynamic decision-making approaches are required to deal with the uncertainty and the plan deviations [15]. According to [7], data-driven decision-making is one of the most critical factors for optimizing disassembly toward being more cost-effective and sustainable. Furthermore, product-individual and automated reactive disassembly planning is required for flexible and semi-automated disassembly systems, which are seen as beneficial for improving the profitability and the sustainability of disassembly [14, 16]. Therefore, data-driven approaches for dynamic and product-individual decision-making in reactive disassembly planning are required.

To address this need, a data-driven approach is presented that supports decision-making in reactive disassembly planning through identifying and selecting solution alternatives, creating a case base, and enabling case-based reasoning (CBR). This paper consists of six sections. Section 2 presents the fundamentals and the challenges of reactive disassembly planning, which are used to derive requirements for reactive planning approaches. In section 3, existing approaches are presented, and research needs are identified. The design of the proposed planning approach is described in section 4. Section 5 describes how the approach enables CBR. The paper concludes with a critical discussion of the presented approach in section 6 and a summary in section 7.

2. Fundamentals and challenges

This section presents the fundamentals and the challenges of decision-making in reactive disassembly planning that are necessary to understand the proposed approach. These include the decision-making tasks (Section 2.1), the decision-making environment (Section 2.2), and the decision-making impact (Section 2.3) of reactive disassembly planning. From the identified challenges, requirements for reactive disassembly planning approaches are derived. According to [12], disassembly planning can be divided into a long-term predictive, a short-term predictive, and a reactive phase. Predictive planning considers a generic product since the conditions of a specific product, as revealed during the disassembly process, are still unknown. The disassembly process can be started with a predictive plan. Still, it should be dynamically adapted during the disassembly process whenever new observations provide information that deviates from the previous assumptions (see Fig. 1). This process of re-establishing a valid plan is known as reactive disassembly planning. [10] It is performed at the operational level and can be described as the interface to disassembly control [11].

2.1. Decision-making tasks

As presented in section 1, reactive disassembly planning must address disposition-, personnel-, equipment-, and product-related deviations from the disassembly plan. This can be achieved through schedule or process adaptations. Schedule adaptations involve assigning individual products to alternative disassembly stations, thereby controlling their flow through the disassembly system. Process adaptations can be achieved by updating the disassembly depth, which refers to the number of components to be disassembled, or by selecting alternative disassembly procedures and tools. [8] Therefore, the decision-making tasks involve determining which adaptations to make and how to combine them to re-establish a valid plan given new information.

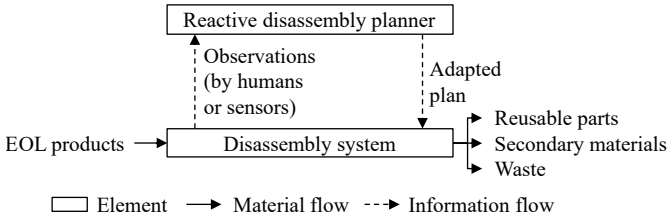


Fig. 1. Concept of a reactive disassembly planner [17]

Reactive disassembly planning searches for alternative disassembly plans by combining schedule and process adaptations. Then, solution alternatives are evaluated regarding the technological feasibility as well as the economic and ecological efficiency, balancing all relevant costs with the possible value generated by the disassembly output. This includes assessing the availability of resources and online scheduling, considering the resulting utilization of the disassembly system. [12] The decision-making tasks, therefore, depend on the individual product [16, 18], its progress in the disassembly process [8, 9], and the state of the resources in the disassembly system [19, 20]. These are all three elements that make up for a disassembly system [21]. This leads to the first challenge: *The decision-making tasks in reactive disassembly planning are complex and individual.* From this challenge, two requirements can be derived. It requires reactive disassembly planning approaches to enable product-individual decision-making (R1) [16] and consider information on products, processes, and resources (R2) [21].

2.2. Decision-making environment

Disassembly involves a wide range of uncertainties, including the product mix and volume, the condition of the product and its components, the duration and probability of success of the disassembly steps, and the demand and the prices for the disassembly output [11]. Due to the information deficit about arriving products, predictive disassembly planning must make comprehensive assumptions [8]. During disassembly, a large amount of product and process information (e.g., valid process times and disassembly costs) is generated and can be

collected by sensors or humans [6]. This leads to a decreasing uncertainty with increasing progress in the disassembly process. Newly acquired information can improve estimates and facilitate the identification of deviations, thereby enhancing decision-making. Combined with the predictive disassembly plan, this information is the input for reactive disassembly planning. [8] Therefore, reactive disassembly planning must deal with uncertainties and newly gained information, make sequential decisions, and enable the disassembly system to adapt. This leads to the second challenge: *The decision-making environment of reactive disassembly planning is uncertain and dynamic.* Three requirements can be derived from this challenge. First, approaches for disassembly planning must consider uncertainty (R3) [15]. Given the uncertainty and dynamic nature of the decision-making environment, fully quantified models cannot be expected to be available in practice [22]. Therefore, approaches for disassembly planning cannot be pre-trained but must accumulate actual data from experience [22]. To gain experience, they must be able to react to new and unknown deviations (R4) [15, 16, 23]. Then, the approaches must exploit the knowledge captured in the accumulated data [22]. Due to the dynamic decision-making environment, this requires the approaches to have the ability to learn quickly and continuously (R5).

2.3. Decision-making impact

Decisions in reactive disassembly planning affect the performance of the disassembly system in multiple aspects. Updating the disassembly depth affects the disassembly output, leading to a change in revenue [11]. Adapting schedules or processes affects the material flow and utilization [8], leading to a shift in cost-effectiveness [7]. In addition, the disassembly output and its condition influence the available circularity strategies for a product's components and, therefore, its ecological footprint [7, 9]. These economic and environmental aspects are interdependent and should be considered simultaneously [23, 19]. This is the third challenge: *The decision-making in reactive disassembly planning must consider its impact on economic and environmental metrics.* Two requirements can be derived from this challenge. First, it requires planning approaches to consider that decisions affect the whole disassembly system. This leads to the requirement for a system perspective (R6) [19, 20]. To quantify the impact, a data-based evaluation of the economic and environmental impact of solution alternatives is required (R7) [16, 23, 24].

3. Existing approaches and research needs

A structured search was conducted in Scopus and Web of Science to review existing approaches for reactive disassembly planning in scientific literature. Search terms were created and connected based on the fundamentals and requirements, such as "adapt* AND disassembly AND planning". Each search term was extended with synonyms by linking them with a logical OR operator. Cluster analysis in VOSviewer was used to exclude research areas outside the scope of the analysis [25]. The search was extended by analyzing the citations and references of identified literature. The identified approaches for reactive disassembly planning are categorized into three categories and discussed below. Table 1 evaluates the identified existing approaches against the specified requirements from section 2.

Planning approaches at the system level (SL in Table 1) address multi-station disassembly systems by considering dependencies between stations in their decision-making. [14] presents an approach that enables the control of multi-station disassembly systems to carry out a given order program as economically as possible. It focuses on developing control measures specific to multi-station disassembly systems and modeling disassembly processes and resources considering uncertainty. In an outlook, a simulation- and data-based learning approach for reactive disassembly planning is suggested to better address uncertainties. [26] focus on the tool and process selection from which control sequences are derived for a multi-station disassembly system. Decisions are made based on a database of process information and sensor-embedded products. [20] present an approach based on Petri net modeling and databases of products, processes, and resources to derive the most profitable disassembly plan. Despite using real-time sensor data, the approach cannot learn based on data. [27] extends this approach with a hybrid Bayesian network that allows reasoning and learning. [28] present an approach for condition-based control of disassembly systems that [29] integrate into a control system that extends from the system level down to the cell level. While a pre-trained reinforcement learning model is used for decision-making at the system level, only selected uncertainties are addressed. [30] focus on minimizing the number of required plan adaptations by proposing an algorithm to determine the most robust disassembly schedule. However, they do not consider process adaptations such as updating the disassembly depth.

Planning approaches at the cell level (CL in Table 1) support the decisions of individual robots or humans without considering any impact on the disassembly system. [31, 32, 33, 34, 35] address cognitive robots in disassembly. Based on information generated by camera vision, the robots can adapt to individual products. For this, [33] and [35] also include learning capabilities. [36] use product-specific digital twin data to recommend circularity strategies for used vehicles. While the digital twin can be updated, the recommendation is based on a fixed logic. [37] propose an approach based on data on used products received from their manufacturer. It considers the product condition and success probabilities of disassembly steps while determining the economically best disassembly sequence. [38, 39, 40, 41, 42] use a fuzzy Petri net model to address uncertainty. A reactive process planner that can learn by adjusting its underlying probability distributions using observed data is developed. [12, 17, 43] introduce approaches for automated disassembly planning using Petri nets and AND/OR graphs, incorporating Bayesian networks to assess probabilities of outcomes. The schedule is adapted based on real-time sensor feedback during disassembly.

Decision-support tools (DS in Table 1) assist with specific aspects of decision-making, but they are not comprehensive

planning approaches. [10] and [44] apply Bayesian networks to combine prior information from a predictive disassembly plan with information gained during the disassembly to support selecting disassembly operations. [45] propose an online prediction model to support process planning based on the condition of a product. [46] and [47] utilize fuzzy reasoning Petri nets to represent products and decision rules for reactive planning considering uncertainty.

Table 1. Evaluation of the identified literature

Category	Publications	R1	R2	R3	R4	R5	R6	R7
SL	[14, 20, 26 – 30]	◑	●	○	○	◑	●	◑
CL	[12, 17, 31 – 43]	●	○	◑	○	◑	○	◑
DS	[10, 44 – 47]	●	○	◑	○	◑	○	◑

● Fulfilled ◑ Partially fulfilled ○ Not fulfilled

SL = Planning approaches at the system level, CL = Planning approaches at the cell level, DS = Decision-support tools

The literature review suggests that current approaches do not meet the specified requirements sufficiently. Four central research needs (N) can be identified to better address the challenges of reactive disassembly planning. Existing approaches with the ability to learn require training data, quantified models, or many trials to make reliable decisions. While existing training data and quantified models are not available in practice, many trials lead to many sub-optimal decisions. This creates the need for an approach that can reliably react to unknown deviations, such as those for new product variants, without requiring many trials (N1). To enable learning based on these initial reactions, the need for an approach to accumulating past decisions into defined data structures can be derived (N2). None of the identified approaches at the system level consider environmental metrics to select solution alternatives. In addition, no identified approach considers the material flow and its effect on utilization and disassembly output. This creates the need for a system-level approach that evaluates solution alternatives based on economic and environmental metrics while considering the impact of material flow on utilization and disassembly output (N3). Finally, current approaches at the system level that are capable of learning are based on reinforcement learning. Reinforcement learning requires relearning whenever decision-making environments abruptly shift [22]. Existing decision-support tools, such as fuzzy reasoning or Bayesian networks, can only gradually adjust their underlying probabilities. Therefore, there is a need for a learning approach that can more quickly adapt to a changing decision-making environment (N4).

4. Data-driven approach for decision-making in reactive disassembly planning

This contribution addresses the identified research needs with a data-driven approach to support decision-making in reactive disassembly planning. The approach is used for the online adaptation of disassembly plans whenever deviations are observed. For this, each deviation is formally described as a problem based on the state of the three elements of a disassembly system (Section 2.1). The solution to each problem is an adapted plan that can be fed back to the disassembly control. The adapted plans can be communicated to humans via dynamic work plans [14] or directly to cognitive robots [16]. Thus, the approach can be implemented in disassembly systems with different levels of automation. This chapter is organized according to the three modules and the procedure of the presented approach, shown in Fig. 2. The steps are explained in detail, following the numeration depicted in Fig. 2. Module I and II originate from the reactive disassembly planning procedure of [12], described in Section 2.1. They form the basis of the approach by enabling it to identify, evaluate, and select adapted plans for unknown problems. Module III connects adapted plans to their corresponding problems and actual results after their application. This structured data is called a case, which is stored in a case base.

4.1. Solution identification (Module I)

Module I identifies solution alternatives for unknown problems, meaning that no similar cases are contained in the case base. The input of the module is the current disassembly plan and the deviations observed by the disassembly control. This requires sufficient transparency of the disassembly system to provide the necessary information, such as the current schedule and the progress of orders. The first step is to formally describe deviations as problems, considering what information to include and how to structure it [48] (Step 1). [49] demonstrates using problem and solution spaces in describing disassembly planning problems. This module applies solution space management to limit the general solution space of reactive disassembly planning to a problem-specific solution space (Step 2). While the general solution space contains all possible measures that reactive disassembly planning can take to adapt plans, the problem-specific solution space only includes those available to react to the given deviation. Then, design of experiments methods are utilized to identify promising solution alternatives from the problem-specific solution space (Step 3). This reduces the total number of experiments [50]. The output of the module is a set of solution alternatives for the given problem.

4.2. Solution selection (Module II)

Module II evaluates the solution alternatives based on their impact on economic and environmental metrics using a material flow simulation model of the disassembly system (Step 4). Simulation can support operational decision-making in uncertain environments at the system level [51]. [16, 52] suggest using such models for data-driven real-time coordination of multi-station disassembly systems and disassembly plan selection. The simulation model is configured based on the problem description. Then, simulation experiments are carried out for each solution alternative to generate data reflecting their impact on the material flow and disassembly output. The generated data is evaluated with regard to disassembly-specific economic and environmental metrics (Step 5). This allows for selecting the

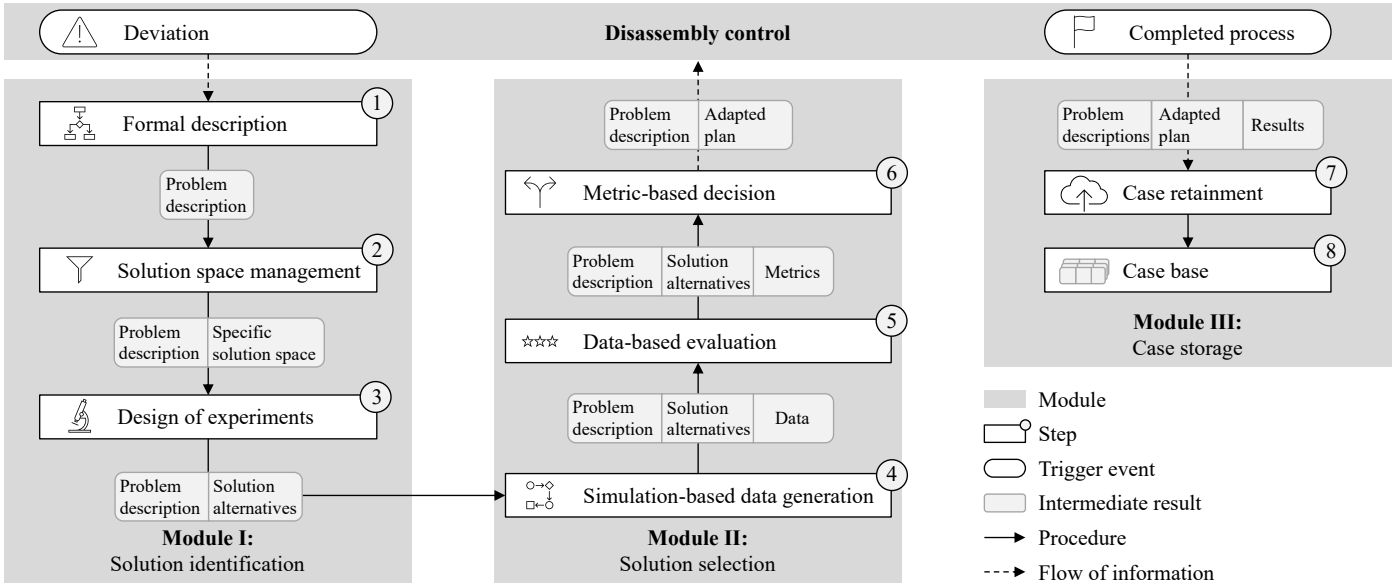


Fig. 2. Modules and procedure of the presented approach

most suitable solution alternative (Step 6). The output of the module is an adapted disassembly plan that is fed back to the disassembly control.

4.3. Case storage (Module III)

Module III contains a case base that stores past cases and is updated each time a disassembly process with deviations is completed so that its actual results are known. Case retainment involves selecting what information from the case to retain, in what form to retain it, how to index the case for later retrieval of similar problems, and how to integrate the new case into the case base (Step 7) [48]. To ensure an effective integration, it is necessary to define how cases should be organized and indexed inside the case base [48] (Step 8). It is also possible to extend the case base by a domain knowledge model, such as an ontology for disassembly [18]. By this, Module III enables the approach to accumulate experience from applying Modules I and II to defined data structures.

5. Utilizing the case storage for case-based reasoning

Applying the presented approach enables the solution of future problems by CBR, assuming that similar cases already exist in the case base (see Fig. 3). This follows approaches such as the ones from [49] and [18], which proposed CBR and case bases for predictive disassembly planning. CBR is an approach that can directly reuse each solved case for all sufficiently similar cases, leading to quick adaptation to changing decision-making environments or new problems. With an increasing number of cases in the case base, the problem-solving ability of a CBR system improves continuously. [18] CBR utilizes the case base created by the presented approach to solve problems whenever similar cases exist in the case base. Therefore, it is a shortcut alternative to applying Modules I and II. This shortcut also requires the formal description of deviations (Step 1 in Fig. 3). For this, the methods from Module I can be used. Then, cases are retrieved from the case base. Case retrieval includes identifying similarity features, the initial matching of similar cases given a similarity threshold, searching for the best match in the initial set, and selecting this best match [48] (Step 2 in Fig. 3). The matching case is then adjusted and reused (Step 3 in Fig. 3). The CBR procedure, therefore, has the same inputs and outputs as the application of Modules I and II. In addition, CBR can incorporate case revision if a proposed solution does not work as expected. The revision process allows the approach to learn from failures by adjusting cases in the case base [48]. In summary, utilizing the case base for CBR allows for faster responses to recurring deviations without applying Modules I and II.

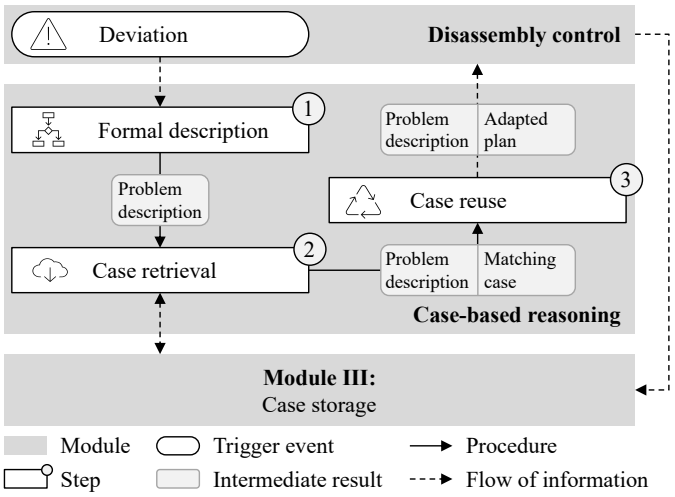


Fig. 3. Intended procedure of CBR for reactive disassembly planning

6. Discussion

Section 2 presents the challenges and the requirements in reactive disassembly planning, which are addressed by the planning approach presented in section 4. Compared to previous works, the proposed approach allows for the product-individual decision-making (R1) for new and unknown deviations (R4) by the automated identification of solution alternatives in Module I. In Module II, the solution alternatives are evaluated and selected based on information on products, processes, and resources (R2) generated by a material flow simulation considering uncertainty (R3). This allows for evaluating their economic and environmental impact (R7) from a system perspective (R6). The modules I and II, therefore, address N1 and N3. Module III accumulates experience into defined data structures by storing past decisions in a case base. This addresses N2. The result of the approach is a continuously increasing case base that enables learning through CBR. CBR can quickly adapt to a changing decision-making environment and continuously improve (R5) with an increasing number of cases in the case base. The approach, therefore, addresses N4 by enabling CBR.

However, the proposed approach also has limitations. Its implementation requires considerable expertise. In addition, it relies on data from the disassembly system. Although this is recognized as a critical prerequisite in research, disassembly systems in practice are still characterized by low data availability [6, 11, 16]. In addition to improving digitization and data collection, synthetic data generation can be utilized. Finally, the approach requires near-real-time performance, which can be a challenge in practical implementation [24]. For recurring problems, this limitation can be addressed by applying CBR, which allows for faster responses without the need for simulation.

7. Summary and future research

This contribution presents a data-driven approach that supports decision-making in reactive disassembly planning by identifying and selecting solution alternatives, creating a case base, and enabling CBR. It allows for the automatic adaptation of disassembly plans in response to plan deviations, considering an adaptation's impact on economic and environmental metrics. Methods for solution space management and design of experiments, as well as a material flow simulation model, are combined to identify and select solution alternatives for unknown deviations. Once a solution alternative is selected, it is linked with its corresponding deviation and stored as a case in a case base. This enables the application of CBR, which can utilize the case base and act as a shortcut to identifying and selecting solution alternatives to accelerate the reaction to recurring plan deviations. Future contributions will focus on the further development and the practical validation of the presented modules. This will include developing methods for the formal description of deviations, solution space management and design of experiments. Furthermore, the automated problem-specific configuration of simulation models will be investigated to then utilize such a model to evaluate solution alternatives. Finally, a case base for reactive disassembly planning problems will be developed. In future research, CBR algorithms applicable to the case base could then be developed. The case base could also be used for exploratory data analysis, e.g., by applying process mining. In addition, the approach could be connected to predictive disassembly planning approaches, improving predictions by giving feedback about occurring deviations.

Acknowledgements

The authors express their gratitude to the Bavarian Research Foundation for their kind support within the project "Prodbot – Intelligente Assistenz in der Produktion" (Grant number: AZ-1518-21).

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